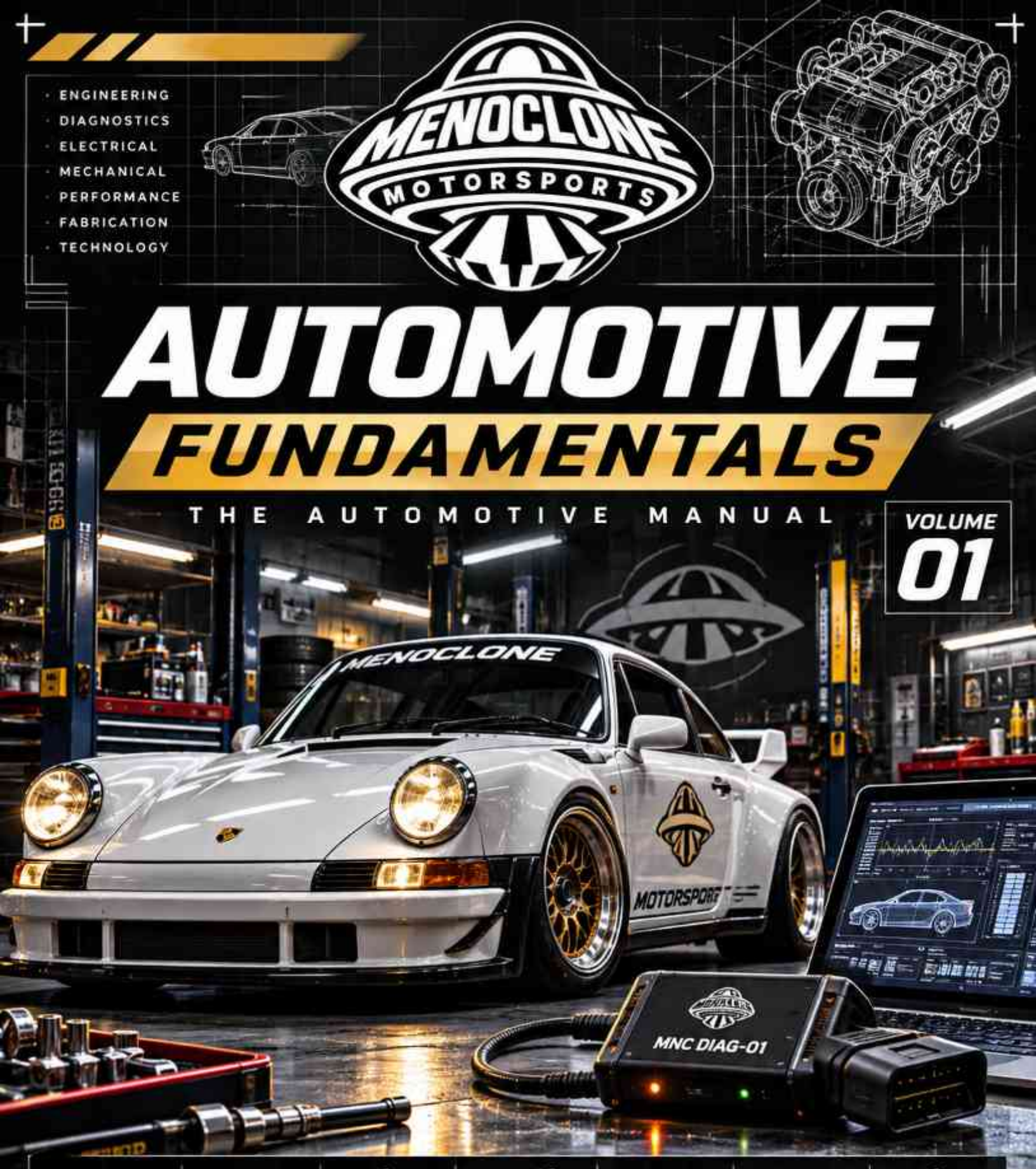
- ENGINEERING
- DIAGNOSTICS
- ELECTRICAL
- MECHANICAL
- PERFORMANCE
- FABRICATION
- TECHNOLOGY

MENOCLONE MOTORSPORTS

AUTOMOTIVE FUNDAMENTALS

THE AUTOMOTIVE MANUAL

VOLUME 01



ENGINE SYSTEMS | ELECTRICAL SYSTEMS | BRAKE SYSTEMS | SUSPENSION & STEERING | OBD II DIAGNOSTICS | TOOLS & SERVICE | ASE PREP

A MENOCLONE MOTORSPORTS EDUCATION SERIES

CHAPTER 02

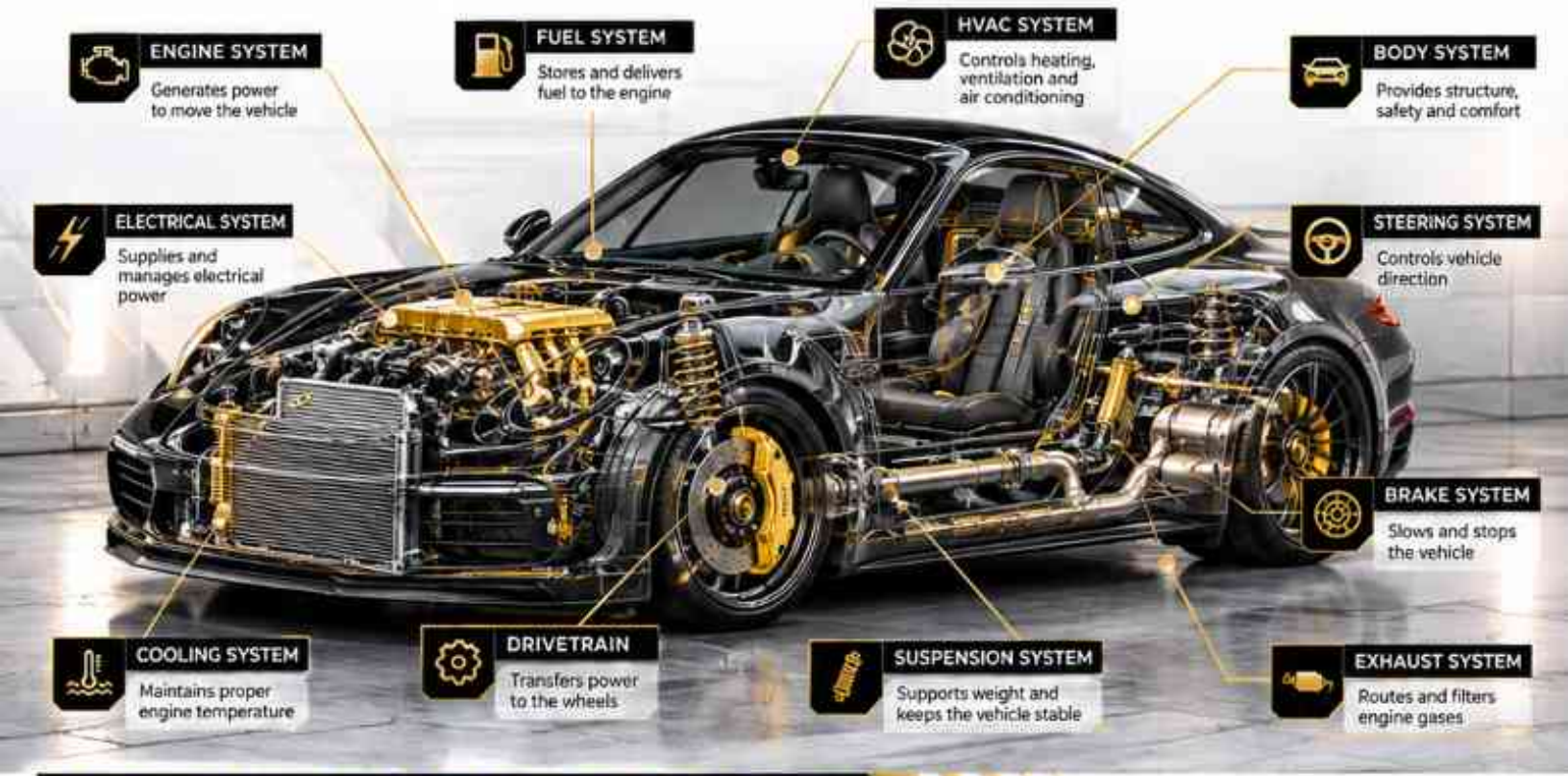
AUTOMOTIVE SYSTEMS

UNDERSTANDING HOW IT ALL WORKS TOGETHER

An automobile is a combination of mechanical, electrical, electronic, and hydraulic systems working together. Understanding how each system functions and interacts is essential for accurate diagnosis, proper maintenance and high performance.

LEARNING OBJECTIVES

01 Identify the main systems in an automobile
02 Understand the purpose of each system
03 Recognize how systems work together
04 Locate major components on a vehicle
05 Build a foundation for advanced diagnostics



THE MAIN AUTOMOTIVE SYSTEMS

1. ENGINE SYSTEM



- Converts fuel to mechanical energy
- Intake, compression, combustion, exhaust
- Includes engine block, pistons, valves, camshaft, crankshaft

2. FUEL SYSTEM



- Stores fuel in the tank
- Delivers fuel at the correct pressure
- Includes fuel pump, filter, injectors, fuel lines and fuel pressure regulator

3. ELECTRICAL SYSTEM



- Supplies 12V power
- Charges the battery
- Starts the engine
- Powers lights, accessories and modules
- Includes battery, alternator, starter, wiring and fuses

4. COOLING SYSTEM



- Maintains proper engine temperature
- Circulates coolant
- Includes radiator, water pump, thermostat, hoses and coolant

5. EXHAUST SYSTEM



- Routes and filters exhaust gases
- Reduces emissions and noise
- Includes exhaust manifold, catalytic converter, resonator and muffler

6. DRIVETRAIN



- Transfers engine power to the wheels
- Includes transmission, driveshaft, differential, axles and CV joints

7. SUSPENSION SYSTEM



- Supports vehicle weight
- Absorbs road impacts
- Keeps tires in contact with the road
- Includes shocks, struts, springs, control arms and bushings

8. BRAKE SYSTEM



- Slows and stops the vehicle
- Includes brake pads, rotors, calipers, master cylinder, brake lines and ABS

9. STEERING SYSTEM



- Controls vehicle direction
- Includes steering wheel, column, rack and pinion, power steering pump and tie rods

10. HVAC SYSTEM



- Controls heating, ventilation and air conditioning
- Includes blower motor, heater core, evaporator, compressor and ducts

11. BODY SYSTEM



- Provides structure and safety
- Supports all systems
- Includes body panels, frame, windows, doors, safety systems and interior components

CHAPTER 02

THE FOUR STROKE ENGINE

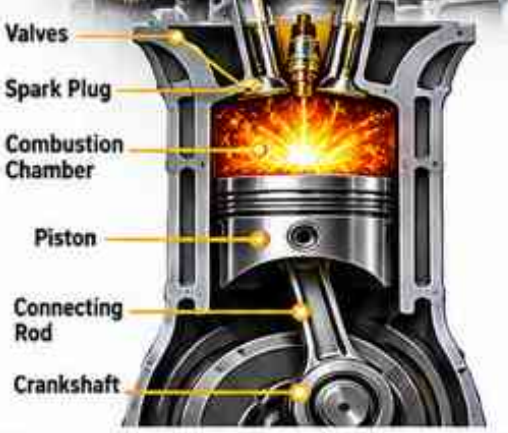
THE HEART OF THE AUTOMOBILE

A four stroke engine is an internal combustion engine that completes one full power cycle in four separate piston movements, called strokes. It converts the energy from fuel and air into mechanical power to move the vehicle.



THE FOUR STROKES

1. **INTAKE**
   Draws in a mix of air and fuel into the cylinder.
2. **COMPRESSION**
   Squeezes the air fuel mix to increase pressure and temperature.
3. **POWER**
   Ignites the compressed mix, creating an explosion that pushes the piston down.
4. **EXHAUST**
   Expels the burned gases out of the cylinder.



KEY COMPONENTS



Piston – Moves up and down inside the cylinder.



Connecting Rod – Links the piston to the crankshaft.



Crankshaft – Converts the piston's up and down motion into rotational power.



Valves – Control the flow of air fuel in and out of the cylinder.



Spark Plug – Creates the spark that ignites the air fuel mix.

THE FOUR STROKE CYCLE

1 INTAKE STROKE



The intake valve opens and the piston moves down, drawing in a mixture of air and fuel.

2 COMPRESSION STROKE



Both valves close and the piston moves up, compressing the air fuel mix.

3 POWER STROKE



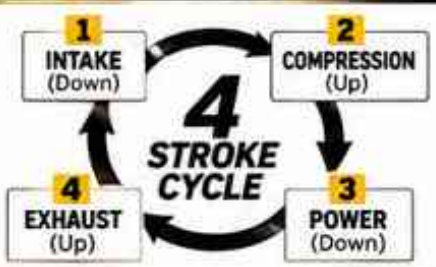
The spark plug fires, igniting the compressed mix. The expansion of the burning gases pushes the piston down, creating power.

4 EXHAUST STROKE



The exhaust valve opens and the piston moves up, pushing the burned gases out of the cylinder.

ENGINE CYCLE OVERVIEW



KEY FACTS

- ✔ Completes one full cycle in four piston strokes (720 degrees of crankshaft rotation).
- ✔ Most common engine design in modern vehicles.
- ✔ More efficient, cleaner and more reliable than two-stroke engines.
- ✔ Used in gasoline, diesel and many performance import engines.



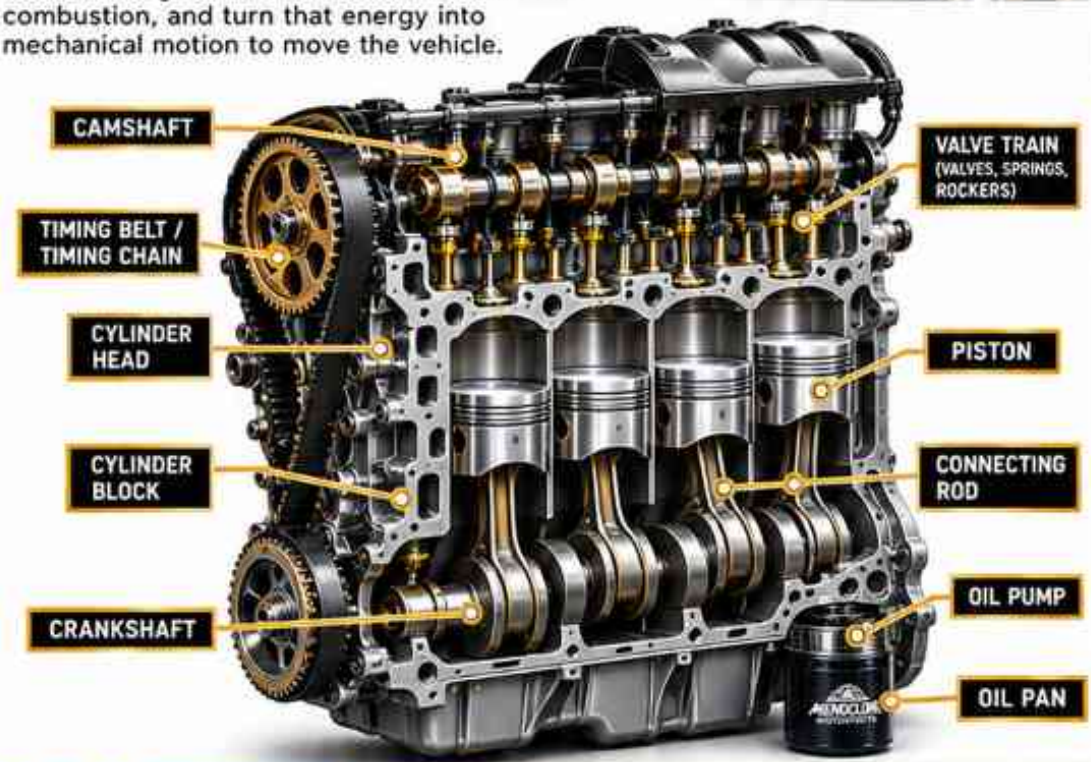
CHAPTER 03

ENGINE COMPONENTS

THE BUILDING BLOCKS OF POWER



An engine is made up of many precision components that work together to mix air and fuel, create combustion, and turn that energy into mechanical motion to move the vehicle.



MAIN ENGINE SYSTEMS

TOP END
Manages air flow, fuel delivery and exhaust flow (cylinder head, valves, camshaft).

BOTTOM END
Converts combustion force into rotational motion (pistons, connecting rods, crankshaft).

AIR & FUEL SYSTEM
Brings in clean air and the correct amount of fuel for combustion (intake, throttle, injectors).

LUBRICATION SYSTEM
Circulates oil to reduce friction, remove heat and prevent wear (oil pump, oil filter, oil pan).

COOLING SYSTEM
Controls engine temperature and prevents overheating (water pump, thermostat, radiator).

KEY ENGINE COMPONENTS & FUNCTIONS

CYLINDER BLOCK



Main structure that houses the cylinders, crankshaft and oil passages.

CYLINDER HEAD



Seals the top of the cylinders and contains the intake and exhaust valves.

PISTON



Moves up and down in the cylinder, compressing the air fuel mix and receiving combustion force.

CONNECTING ROD



Links the piston to the crankshaft and transfers the force of combustion.

CRANKSHAFT



Converts the piston's up and down motion into rotational motion to power the vehicle.

CAMSHAFT



Opens and closes the intake and exhaust valves at the correct time.

VALVES



Control the flow of air fuel into the cylinder and the flow of exhaust gases out.

TIMING BELT / CHAIN



Keeps the camshaft and crankshaft in sync for proper engine timing.

OIL PUMP



Draws oil from the oil pan and circulates it through the engine.

OIL PAN



Holds the engine oil and helps cool it before it circulates again.

COMPONENT LAYOUT (CROSS SECTION)



KEY TAKEAWAYS

- ✔ Each engine component has a specific job.
- ✔ The top end controls air, fuel and exhaust flow.
- ✔ The bottom end converts combustion into rotation.
- ✔ Lubrication and cooling keep the engine reliable.
- ✔ All components work together for power, efficiency and long life.



CHAPTER 04

AUTOMOTIVE TOOLS

THE ESSENTIAL TOOLS OF THE TRADE

Automotive tools are the equipment used to inspect, diagnose, service and repair vehicle systems. Using the right tool for the job improves safety, accuracy and efficiency, especially when working on modern import vehicles.



HAND TOOLS

Basic tools used for general service, assembly and disassembly.



WRENCHES
Loosen and tighten nuts and bolts.



RATCHETS & SOCKETS
Provide leverage and fit various fasteners.



SCREWDRIVERS
Remove and install screws and small parts.



PLIERS
Grip, hold, bend and cut wire or components.



HAMMERS
Used for light striking, assembly and removal.

POWER TOOLS

Motorized tools that save time and increase productivity.



IMPACT WRENCH
Quickly removes and installs lug nuts and high torque fasteners.



DRILL / DRIVER
Drills holes and drives screws for installation and fabrication.



DIE GRINDER
Used for grinding, cutting and surface preparation.

DIAGNOSTIC TOOLS

Electronic tools used to read, test and troubleshoot vehicle systems.



OBD II SCAN TOOL
Reads diagnostic trouble codes (DTCs), live data and system information.



MULTIMETER
Measures voltage, current and resistance (Ohm's Law).



TEST LIGHT
Quickly checks for power, ground and circuit operation.

SPECIALTY TOOLS

Purpose built tools for specific automotive tasks, especially for import vehicles.



TORQUE WRENCH
Tightens fasteners to the manufacturer's specified torque.



OIL FILTER WRENCH
Removes and installs oil filters.



BRAKE CALIPER TOOL
Compresses caliper pistons for brake service.



COMPRESSION TESTER
Measures engine cylinder compression (engine health).



VACUUM / BOOST GAUGE
Checks engine vacuum and boost for performance and diagnostics.



FUEL PRESSURE TESTER
Measures fuel system pressure for proper operation.



TOOL SAFETY & BEST PRACTICES

- Use the correct tool for the job.
- Inspect tools for damage before use.
- Keep your work area clean and organized.
- Follow the vehicle manufacturer's service procedures.
- Wear proper personal protective equipment (PPE).

PERSONAL PROTECTIVE EQUIPMENT (PPE)



SAFETY GLASSES
(EYE PROTECTION)



MECHANIC GLOVES
(HAND PROTECTION)



EAR PROTECTION
(NOISE CONTROL)



RESPIRATOR / MASK
(DUST & CHEMICALS)



SAFETY BOOTS
(FOOT PROTECTION)

TOOL MAINTENANCE

- Clean tools after each use.
- Keep moving parts lubricated.
- Store tools in a dry, organized toolbox.
- Replace worn or damaged tools.

CHAPTER 05

ELECTRICAL FUNDAMENTALS

THE FLOW OF POWER IN THE VEHICLE

Automotive electrical systems use a 12 volt direct current (DC) system to supply, control and distribute electrical power to start the engine, charge the battery and operate vehicle components such as lights, sensors, modules and accessories.



KEY ELECTRICAL TERMS

VOLTAGE (V)	Electrical pressure that pushes electrons through a circuit (measured in volts).
CURRENT (I)	The flow of electrons (measured in amperes or amps).
RESISTANCE (R)	Opposition to the flow of current (measured in ohms Ω).
POWER (P)	The rate of electrical work (measured in watts).
DC (Direct Current)	Electrical flow in one direction (used in vehicles, typically 12V).
GROUND	The return path for current (usually connected to the vehicle chassis).

THE AUTOMOTIVE ELECTRICAL SYSTEM



OHM'S LAW

The relationship between voltage, current and resistance in an electrical circuit.

$$V = I \times R$$

V = Voltage (volts)
I = Current (amps)
R = Resistance (ohms)



USEFUL FORMULAS

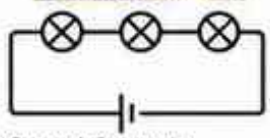
$$V = I \times R$$

$$I = V \div R$$

$$R = V \div I$$

CIRCUITS: SERIES vs. PARALLEL

SERIES CIRCUIT



- One path for current.
- Same current through all loads.
- Total resistance increases.
- If one load fails, all turn off.

PARALLEL CIRCUIT



- Multiple paths for current.
- Same voltage at each load.
- Total resistance decreases.
- If one load fails, others stay on.

KEY ELECTRICAL COMPONENTS



BATTERY
Stores chemical energy and supplies 12V DC power.



ALTERNATOR
Generates electrical power and recharges the battery (≈13.5–14.8V).



STARTER
Uses electrical power to crank the engine and start combustion.

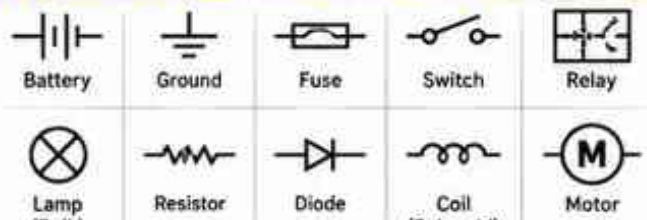


RELAY
An electrically controlled switch that powers higher current circuits.



FUSE
A safety device that opens the circuit when current is too high.

COMMON ELECTRICAL SYMBOLS





USING A MULTIMETER

1. **VOLTAGE (V)**
   Check battery voltage or circuit supply (DC volts, e.g. 12.6V).
2. **CONTINUITY (•))**
   Verify a complete circuit (beep = good path).
3. **RESISTANCE (Ω)**
   Measure resistance of a component (with power off).
4. **CURRENT (A)**
   Measure electrical flow (use in series with the circuit) and correct meter range).

KEY TAKEAWAYS

- ✔ Automotive systems use 12 volt DC power.
- ✔ Voltage, current and resistance are related by Ohm's Law.
- ✔ Circuits can be series, parallel or a combination.
- ✔ Fuses and relays protect and control circuits.
- ✔ Use a multimeter to test, diagnose and verify circuits safely.

CHAPTER 06

BATTERIES AND CHARGING SYSTEMS

STORAGE POWER CHARGE CONTROL

The battery stores electrical energy to start the engine and power vehicle systems, while the charging system (alternator, regulator and related components) keeps the battery charged and supplies power while the engine runs.



THE AUTOMOTIVE BATTERY

A rechargeable 12 volt battery that stores chemical energy and converts it to electrical energy (DC) using a chemical reaction.



BATTERY SPECS (TYPICAL)

Voltage	12 Volt (DC)
Cells	6 (≈ 2.1V each)
Capacity	40 – 100 Ah (Amp-Hours)
Type	Lead Acid (AGM, EFB, Conventional)

THE CHARGING SYSTEM

Generates electrical power, recharges the battery and keeps system voltage at the correct level (typically 13.5 – 14.8 volts).



ALTERNATOR (CUTAWAY VIEW)

HOW THE CHARGING SYSTEM WORKS



1 BATTERY
Supplies initial power to start the engine.

2 ALTERNATOR
Generates AC power and converts it to DC (≈ 13.5 – 14.8V).

3 VOLTAGE REGULATOR
Keeps the charging voltage at a safe and consistent level.

4 CHARGING INDICATOR
Shows system status (ON = problem, OFF = charging normally).

CHARGING SYSTEM COMPONENTS



ALTERNATOR – Generates electrical power and charges the battery.

VOLTAGE REGULATOR – Controls the output voltage (≈ 13.5 – 14.8V).

RECTIFIER (DIODES) – Converts the alternator's AC output to DC power.

DRIVE BELT – Spins the alternator from the engine (serpentine or V-belt).

BATTERY TYPES



CONVENTIONAL (Standard)
Affordable and common (maintenance may be required).

AGM (Absorbed Glass Mat)
Sealed, vibration resistant and ideal for performance and import vehicles.

EFB (Enhanced Flooded Battery)
Stronger, longer life and made for vehicles with start stop systems.

VOLTAGE READINGS (GUIDE)

CONDITION	VOLTAGE (DC)
Battery at rest (engine off)	12.4 – 12.7 V
Low or discharged battery	Below 12.4 V
Charging (engine running)	13.5 – 14.8 V
Overcharging (problem)	Above 14.8 V

TESTING PROCEDURES

1 RESTING VOLTAGE TEST
Engine off for at least 30 minutes. Measure at battery terminals (≈ 12.4 – 12.7V).

2 CHARGING VOLTAGE TEST
Start the engine and measure at the battery. Should read 13.5 – 14.8V.

3 ALTERNATOR OUTPUT TEST
Increase engine speed to ≈ 2,000 RPM. Voltage should stay stable within 13.5 – 14.8V.



COMMON ISSUES & SYMPTOMS

ISSUE	SYMPTOMS
Weak / Bad Battery	Slow crank, no start, low voltage (≤ 12.4V).
Failing Alternator	Battery light on, dim lights, low charging voltage (< 13.5V).
Worn / Loose Drive Belt	Squealing noise, charging problems, low voltage.
Faulty Voltage Regulator	Overcharging (> 14.8V) or undercharging (< 13.5V).
Poor Connections	Intermittent charging, warning light on and off, voltage fluctuations.



KEY TAKEAWAYS

- The battery stores 12 volt DC electrical energy for starting and vehicle operation.
- The alternator recharges the battery and supplies power while the engine runs.
- Normal charging voltage is 13.5 – 14.8 volts.
- Keep the battery, terminals and drive belt in good condition.
- Test regularly with a multimeter to catch problems early.

CHAPTER 07

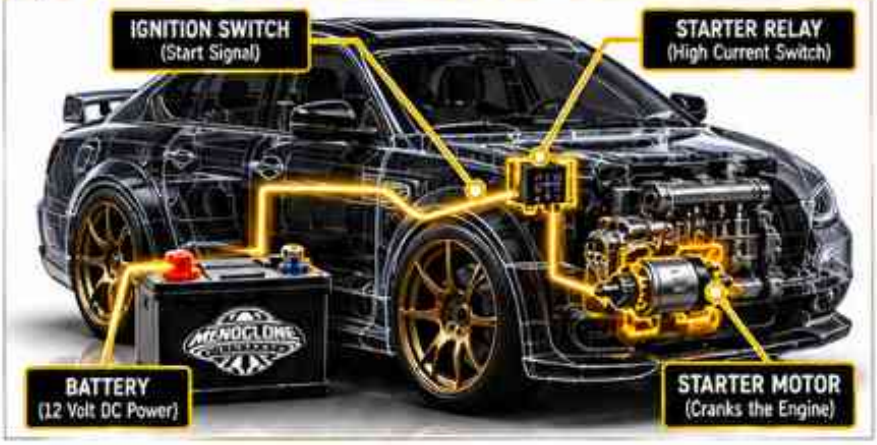
STARTING SYSTEMS

CRANKING POWER TO START THE ENGINE

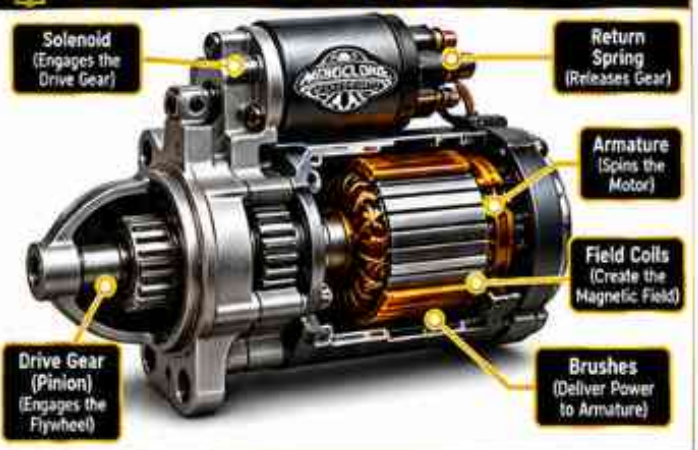
The starting system uses electrical power from the battery to turn the engine's crankshaft fast enough for the engine to start and run on its own.



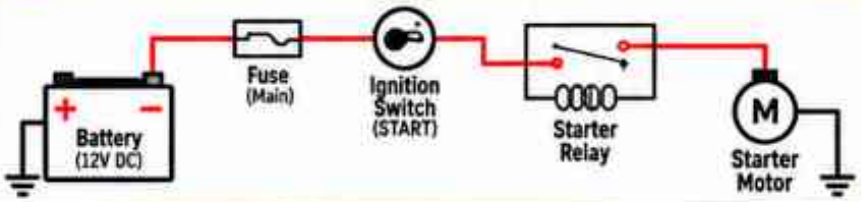
THE STARTING SYSTEM OVERVIEW



THE STARTER MOTOR (CUTAWAY VIEW)



STARTING SYSTEM CIRCUIT (BASIC)



KEY COMPONENTS & FUNCTIONS



BATTERY – Supplies 12 volts DC to the starting system.

IGNITION SWITCH – Sends a start signal to the starter relay.

STARTER RELAY – A high current switch that sends battery power to the starter motor.

STARTER SOLENOID – Moves the drive gear (pinion) into the flywheel and closes the high current circuit.

STARTER MOTOR – An electric motor that turns the engine's flywheel to crank the engine.

FLYWHEEL / RING GEAR – A gear attached to the engine that the starter gear engages to spin the engine.

HOW THE STARTING SYSTEM WORKS



1 Start Command
You turn the key or press the start button, sending a signal to the starter relay.

2 Relay Activates
The starter relay closes, allowing high current from the battery to flow to the solenoid.

3 Solenoid Engages
The solenoid moves the drive gear forward to mesh with the flywheel and powers the starter motor.

4 Engine Cranks
The starter motor spins the flywheel, turning the engine until it starts and runs on its own.

VOLTAGE & CURRENT SPECIFICATIONS (TYPICAL – IMPORT VEHICLES)

System Voltage	12 Volt DC
Cranking Voltage (Minimum)	≥ 9.6 Volts
Cranking Current (Typical)	80 – 250 Amps (varies by engine size)
Starter Relay Coil Voltage	12 Volts
Solenoid Pull-In Current	20 – 40 Amps (brief)

COMMON PROBLEMS & SYMPTOMS

SYMPTOM	POSSIBLE CAUSE
No crank, no sound	Dead battery, bad connection, faulty relay or ignition switch.
Clicking sound only	Low battery, weak connections or failing starter.
Slow crank	Weak battery, high resistance or worn starter motor.
Intermittent start	Failing relay, ignition switch or worn solenoid.
Grinding noise	Damaged drive gear or flywheel (ring gear) teeth.

BASIC DIAGNOSTIC STEPS

1. Check battery voltage (≥ 12.4 V at rest).
2. Check voltage at the starter while cranking (≥ 9.6 V).
3. Test the starter relay (listen or swap with a known good relay).
4. Check power and ground connections (for corrosion or looseness).
5. If power is present and it still won't crank, test the starter motor and solenoid.

KEY TAKEAWAYS

- ✔ The starting system uses the battery, ignition switch, relay, solenoid and starter motor to crank the engine.
- ✔ A healthy battery and clean, tight connections are critical for reliable starting.
- ✔ The starter motor turns the flywheel, which rotates the engine until it starts.
- ✔ Use a multimeter and a step by step process to quickly find starting system problems.

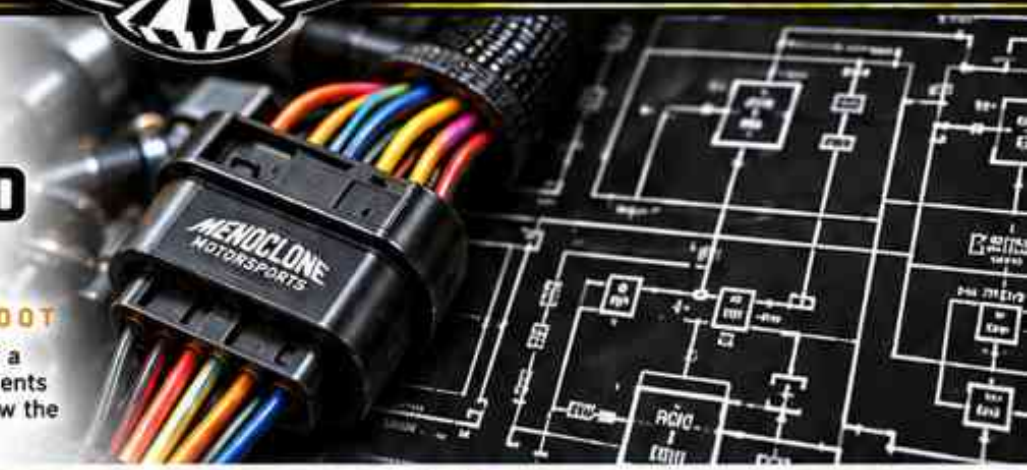


CHAPTER 08

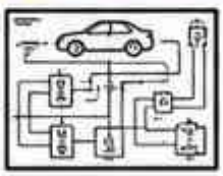
WIRING DIAGRAMS AND SCHEMATICS

READ UNDERSTAND TROUBLESHOOT

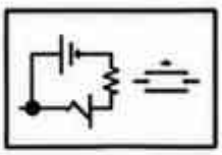
Wiring diagrams and schematics are visual maps of a vehicle's electrical system. They show how components are connected, how power and ground flow, and how the system is designed to operate.



WHAT ARE THEY?



A wiring diagram is a pictorial map that shows the components, wires, connectors, and their relationships in a circuit.



A schematic is a simplified electrical diagram that shows how a circuit works using standard electrical symbols.

TYPES OF DIAGRAMS

SYSTEM WIRING DIAGRAM
Shows the complete wiring for a specific system (ex. starting, charging, lighting, etc.).

CIRCUIT SCHEMATIC
A simplified layout that shows electrical flow and component operation.

CONNECTOR DIAGRAM
Shows the connector shape, pin numbers, and wire colors for each component.

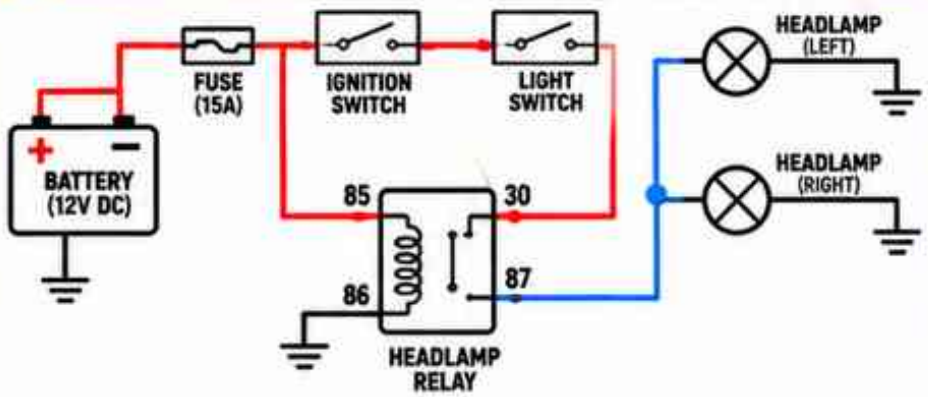
COMMON ELECTRICAL SYMBOLS

Battery	Ground	Fuse	Switch	Relay
Resistor	Diode	Lamp (Bulb)	Motor	Solenoid
Capacitor	Coil (Inductor)	Connector	Splice (Junction)	Crossing (No Connection)

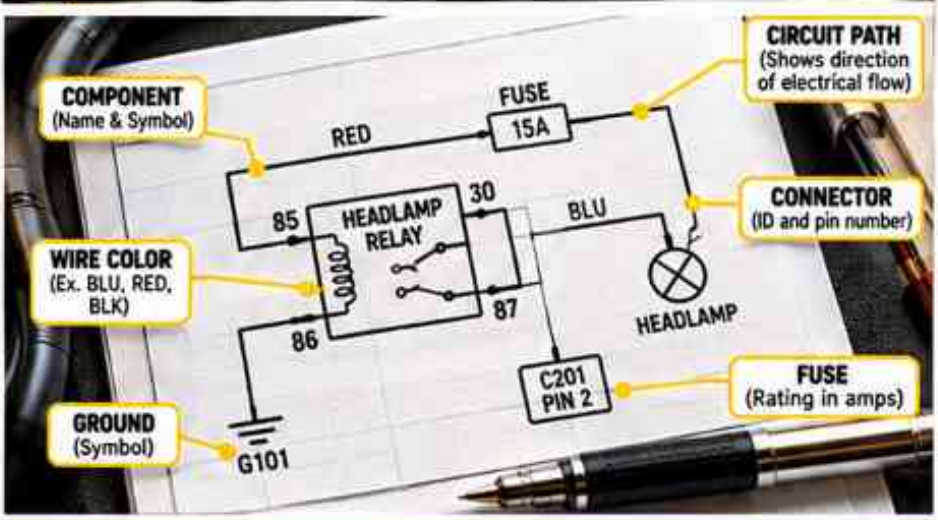
AUTOMOTIVE WIRE COLOR CODE (COMMON)

COLOR	TYPICAL USE
Red	Battery power (constant 12V)
Yellow	Ignition switched power (key ON)
Black	Ground (chassis or chassis ground)
Brown	Accessory power (radio, modules)
Blue	Headlights / lighting circuits
Green	Sensors and signal circuits
White	Control and trigger signals
Orange	Illumination / dash lighting
Purple	Data and communication (CAN, LIN, etc.)
Gray	Low current control circuits

SAMPLE WIRING DIAGRAM – LIGHTING CIRCUIT (SIMPLIFIED)



HOW TO READ A WIRING DIAGRAM



READING STEPS

1. Identify the system (ex. lighting, starting).
2. Find the power source (battery or ignition feed).
3. Follow the circuit path through fuses, switches, relays, and components.
4. Check wire colors, connector IDs, and pin numbers.
5. Locate the ground(s) and confirm the return path.

KEY TAKEAWAYS

- ✓ Wiring diagrams show how components are connected and powered.
- ✓ Learn and recognize standard electrical symbols.
- ✓ Use wire color codes and connector IDs to trace circuits.
- ✓ Diagrams are essential for troubleshooting electrical problems.
- ✓ Always verify the circuit with a multimeter before repair.

CHAPTER 09

SENSORS AND ACTUATORS

THE INPUTS AND OUTPUTS OF THE ECU

Sensors monitor what's happening in the vehicle and send information to the ECU (Engine Control Unit). Actuators receive commands from the ECU and perform an action, such as controlling fuel, air, ignition or other systems.



THE BIG PICTURE

SENSORS (Input)
Measure engine, driving, and environment conditions.

→

ECU / PCM (Processor)
Reads sensor data, makes decisions, and controls the vehicle.

→

ACTUATORS (Output)
Carry out ECU commands (ex. open, close, move, spray, or turn on).

COMMON SENSOR EXAMPLES (INPUTS)



MAF SENSOR
Measures the amount of air entering the engine.



MAP SENSOR
Measures intake manifold pressure (air load).



TPS (Throttle Position Sensor)
Reads the throttle valve position.



ECT SENSOR (Engine Coolant Temp)
Measures engine coolant temperature.



O2 SENSOR (Oxygen Sensor)
Measures oxygen in the exhaust to control the air fuel ratio.

COMMON ACTUATOR EXAMPLES (OUTPUTS)



FUEL INJECTOR
Sprays a precise amount of fuel into the engine.



IAC VALVE (Idle Air Control)
Controls engine idle speed by adjusting air flow.



EGR VALVE (Exhaust Gas Recirculation)
Re-circulates exhaust gases to reduce emissions (NOx).



VVT SOLENOID (Variable Valve Timing)
Adjusts cam timing for performance and efficiency.



EVAP PURGE VALVE
Controls fuel vapors from the fuel tank to the engine intake.

SENSORS AND ACTUATORS IN THE VEHICLE



HOW THEY WORK TOGETHER (EXAMPLE)

Air Fuel Control (Closed Loop System)



1 **O2 Sensor** measures exhaust oxygen levels.

→



2 **ECU** adjusts the air fuel ratio (using the data).

→



3 **Fuel Injector** changes the amount of fuel to keep the mix correct (≈ 14.7:1).

KEY SENSOR CATEGORIES

Temperature Sensors	Monitor temperature (ECT, IAT, ATF).
Pressure Sensors	Measure air, fuel or oil pressure (MAP, Fuel Pressure).
Air / Flow Sensors	Measure air flow or intake load (MAF, MAP).
Position Sensors	Detect component position (TPS, Cam, Crank).
Exhaust / Emissions Sensors	Monitor exhaust gases and emissions (O_2, NOx).

DIAGNOSTIC TIPS



- ✓ Use a scan tool to check sensor readings (live data).
- ✓ Compare readings to manufacturer specs.
- ✓ Look for out of range or no signal faults.
- ✓ Test the sensor circuit (power, ground and signal).
- ✓ Use actuator tests (bi directional control) when available.

KEY TAKEAWAYS

- ✓ Sensors are the input (they tell the ECU what's happening).
- ✓ Actuators are the output (they carry out the ECU's commands).
- ✓ The ECU uses sensor data to control actuators and keep the engine running efficiently.
- ✓ Common sensors include MAF, MAP, TPS, ECT and O_2.
- ✓ Common actuators include fuel injectors, IAC, EGR, VVT and EVAP.
- ✓ Understanding sensor and actuator operation is key to diagnosing modern import vehicles.

ASE CERTIFICATION SERIES

ASE A6 ELECTRICAL SYSTEMS

DIAGNOSIS SERVICE REPAIR

A focused guide for the ASE A6 Electrical / Electronic Systems certification. Covers the theory, diagnosis, service and repair of automotive electrical systems using industry standards and real world testing.



WHAT IS ASE A6?

The ASE A6 Electrical / Electronic Systems exam tests your knowledge and skills in diagnosing, servicing and repairing automotive electrical and electronic systems (12 volt and related circuits).

A6 TOPIC AREAS (EXAM DOMAINS)

A1	General Electrical Theory
A2	Circuit Diagnosis & Repair
A3	Starting System
A4	Charging System
A5	Batteries
A6	Lighting & Accessory Circuits
A7	Electronic Circuits & Components
A8	Vehicle Communication & Controls

KEY SYSTEMS COVERED

- Starting System
- Charging System
- Battery Service
- Lighting Systems
- Relays & Switches
- Sensors & Inputs
- ECU / Control Modules
- CAN / Vehicle Network Basics

CORE A6 SYSTEM OVERVIEW

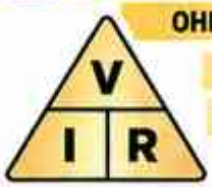


A6 DIAGNOSTIC PROCESS

1. **VERIFY THE CONCERN**
   Understand the customer complaint and perform a visual check.
2. **CHECK POWER, GROUND & FUSES**
   Confirm battery voltage, ground integrity and fuse / relay operation.
3. **TEST THE CIRCUIT**
   Use a multimeter, test light or scan tool to check voltage, continuity and resistance.
4. **ANALYZE INPUTS & OUTPUTS**
   Review sensor data, control signals and module responses.
5. **CONFIRM THE REPAIR**
   Repair the fault, clear codes (if needed) and verify proper operation.

ESSENTIAL FORMULAS & TESTS

OHM'S LAW



$V = I \times R$

$I = V \div R$

$R = V \div I$

V = Voltage (volts)
I = Current (amps)
R = Resistance (ohms)

KEY TESTS (A6)

- ✓ Battery Load Test
- ✓ Charging Voltage Test (13.5 – 14.8 V)
- ✓ Voltage Drop Test (< 0.5 V typical)
- ✓ Continuity Test
- ✓ Resistance Test
- ✓ Current Draw (Parasitic) Test

COMMON A6 TOOLS



- **Multimeter** (V, Ω, A)
- **Scan Tool** (DTCs & Live Data)
- **Clamp Meter** (Current Draw)
- **Test Light** (Circuit Power)
- **Battery Load Tester** (Battery Health)

COMMON A6 FAULTS & SYMPTOMS

SYSTEM / COMPONENT	SYMPTOM (EXAMPLE)
Battery	Slow crank, no start, low voltage
Charging System	Battery light on, nlow charging voltage (< 13.5 V)
Starting System	Click, no crank, intermittent start
Lighting Circuits	Inoperative lights, dim lights, blown fuses
Relays / Switches	Component not operating, no power
Control Modules	Communication codes, sensor or output faults

STUDY & EXAM TIPS

- ✓ Know the operation and testing of starting, charging and battery systems.
- ✓ Be able to read and interpret wiring diagrams.
- ✓ Understand series, parallel and combination circuits.
- ✓ Practice with a multimeter and scan tool.
- ✓ Review common sensor and module functions.
- ✓ Use ASE practice tests to build exam confidence.

KEY TAKEAWAYS

- ✓ ASE A6 focuses on real world electrical diagnosis and repair.
- ✓ Master the core systems: starting, charging, battery, lighting and control circuits.
- ✓ Be skilled in using test equipment and wiring diagrams.
- ✓ Understand how sensors, modules and networks work together.
- ✓ Practice diagnostics to acurately find and fix the fault.

CHAPTER 11

PRACTICAL TROUBLESHOOTING

FIND TEST ISOLATE REPAIR VERIFY

A real world guide to diagnosing and fixing automotive problems using a systematic process, hands on testing and proven methods used by professional technicians and ASE certified repair standards.



THE TROUBLESHOOTING PROCESS

1. **VERIFY THE CONCERN** — Confirm the customer complaint and reproduce the problem.
2. **GATHER INFORMATION** — Check for DTCs, symptoms, history and any recent repairs.
3. **INSPECT VISUALLY** — Look for obvious issues such as loose connectors, damage, leaks or corrosion.
4. **TEST AND MEASURE** — Use the correct tools (multimeter, scan tool, test light) to check voltage, ground, resistance and signals.
5. **ISOLATE THE FAULT** — Narrow down the problem to a specific circuit, component or system.
6. **REPAIR THE ISSUE** — Replace, repair or adjust the faulty part using proper procedures.
7. **VERIFY THE REPAIR** — Clear DTCs, retest the system and confirm the concern is resolved.

DIAGNOSTIC TOOLKIT (ESSENTIALS)



Scan Tool (DTCs, Live Data, Active Tests)



Multimeter (Voltage, Resistance, Continuity)



Test Light (Quick Power / Ground Check)



Clamp Meter (Current Draw / Parasitic Test)



Battery Load Tester (Battery Health / Cranking Power)



Oscilloscope (Sensor, Signal and Waveform Analysis)

COMMON SYMPTOMS AND POSSIBLE AREAS

- **No Start / Crank** — Battery, starting system, ignition, fuel or immobilizer.
- **Weak Crank** — Battery, connections, parasitic draw, starter.
- **No Power / Dead Circuit** — Fuse, relay, wiring, ground or control module.
- **Lights Not Working** — Bulb, fuse, switch, relay or ground.
- **Rough Idle / Stalls** — Air, fuel, ignition, sensor or vacuum leak.
- **Check Engine Light (MIL)** — Read DTCs, inspect related sensors and circuits.
- **Overheating** — Coolant level, thermostat, fan, relay or temperature sensor.

ELECTRICAL TESTING BASICS

Voltage Test (Power Supply)	Continuity Test (Wiring Check)	Resistance Test (Component Check)	Current Draw Test (Parasitic / Load)
Check for correct voltage (ex. 12 V at battery, 5 V at sensor).	Verify a complete circuit (low resistance, near 0 ohms).	Compare the measured resistance to the manufacturer specification.	Measure electrical current (amperage) to find excess draw or a faulty component.

DIAGNOSTIC FLOW EXAMPLE (Check Engine Light On)



COMMON DTC CATEGORIES

CODE TYPE	EXAMPLE
P – Powertrain	Engine, transmission, emissions (Ex. P0101 – MAF Sensor Range)
B – Body	Interior systems (Ex. B1245 – Door Circuit)
C – Chassis	Brakes, ABS, traction control (Ex. C0035 – Wheel Speed Sensor)
U – Network	Communication between modules (Ex. U0100 – Lost Communication with ECM)

REAL WORLD EXAMPLE (No Start – Diagnostic Approach)



1. Verify the no start condition.
2. Check battery voltage (≥ 12.4 V at rest).
3. Perform a voltage drop test (on positive and ground).
4. Test the starter control circuit (ignition switch to starter relay).
5. Test the starter (power and ground at the motor).
6. Repair the failed component (ex. relay, wiring or starter).
7. Recheck and confirm the engine starts normally.

TIPS FOR SUCCESS

- ✓ Follow a step by step process.
- ✓ Use a wiring diagram or service manual.
- ✓ Check for Technical Service Bulletins (TSBs).
- ✓ Confirm power, ground and signal at the component.
- ✓ Compare readings to manufacturer specs.
- ✓ Make one change at a time.
- ✓ Always verify the repair with a full system test.

KEY TAKEAWAYS

- ✓ A structured process saves time and avoids guesswork.
- ✓ Use the right tools and information.
- ✓ Diagnose by testing, not just replacing parts.
- ✓ Understand the system, circuit and component operation.
- ✓ Confirm the repair and return the vehicle with confidence.

FINAL ASSESSMENT

AUTOMOTIVE ELECTRICAL SYSTEMS MASTER REVIEW

ASE A6 COMPREHENSIVE EVALUATION

A complete final assessment covering all major topics from the Automotive Electrical Systems series. Test your knowledge, diagnose real world scenarios, and prepare for ASE A6 certification and on the job success.



COURSE TOPICS COVERED (REVIEW)

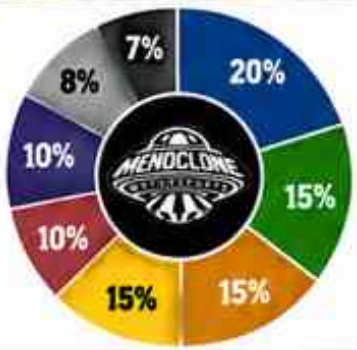
✓ Engine Components	Engine systems and electrical integration
✓ Automotive Tools	Identification, safe use and testing
✓ Electrical Fundamentals	Voltage, current, resistance and Ohm's Law
✓ Batteries & Charging Systems	Battery operation, testing and alternator charging
✓ Starting Systems	Starter, solenoid, circuit operation and diagnosis
✓ Wiring Diagrams & Schematics	Reading, interpreting and tracing electrical circuits
✓ Sensors & Actuators	Inputs, outputs and ECU control
✓ ASE A6 Electrical Systems	Exam standards, diagnosis, service and repair
✓ Practical Troubleshooting	Step by step diagnosis and repair strategies

FINAL ASSESSMENT FORMAT

50	10	5	5
Multiple Choice Questions (Knowledge & Theory)	Diagnostic Scenarios (Real World Problems)	Wiring Diagram Tasks (Read & Interpret)	Service & Test Procedures (Step by Step)

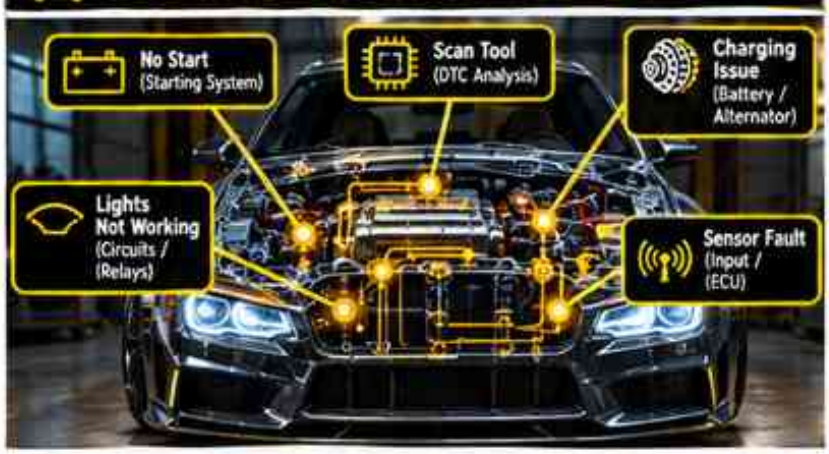
TOTAL: 70 POINTS | PASSING SCORE: 70% (49 / 70)

ASSESSMENT COVERAGE BY TOPIC



- 20% Electrical Fundamentals
- 15% Batteries & Charging Systems
- 15% Starting Systems
- 15% Wiring Diagrams & Schematics
- 10% Sensors & Actuators
- 10% Practical Troubleshooting
- 8% Engine Components & Tools
- 7% ASE A6 Exam Knowledge

REAL WORLD DIAGNOSTIC CHALLENGE



SAMPLE FINAL ASSESSMENT QUESTIONS

1. A battery tests at 12.2 volts with the engine off. What does this indicate?
2. What is the correct voltage range for a charging system at idle?
3. A starter clicks but does not crank. What are two likely causes?
4. What does a dropped voltage test help you find?
5. In a wiring diagram, what does a dashed line typically represent?

PRACTICAL TASKS (HANDS ON OR SIMULATED)



Test Battery Condition & State of Charge



Verify Charging System Output (Alternator Test)



Diagnose a No Start Condition (Starter Circuit)

FINAL CHECKLIST

- ✓ Review all chapter study guides
- ✓ Complete practice quizzes for each topic
- ✓ Work through the diagnostic scenarios
- ✓ Practice reading at least 5 wiring diagrams
- ✓ Perform hands on tests with a multimeter
- ✓ Score 70% or higher on the final assessment
- ✓ Identify weak areas and review as needed
- ✓ Be confident in A6 scope and real world application

CERTIFICATION GOAL

ASE CERTIFIED

ASE A6 ELECTRICAL / ELECTRONIC SYSTEMS

✓ BE PREPARED
✓ TEST READY
✓ WORK READY